

Petroleum Refining
Design Basis Accident — Military Action
DBA-OIL-FC1 — Facility Class Document
Issued Under: DBA-OIL Sector Framework

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Preface

The petroleum refinery is the most consequential single facility class in the petroleum sector. A refinery is not merely a large industrial plant — it is the transformation node through which crude oil becomes the fuel products on which transportation, military logistics, agriculture, and manufacturing depend. When a refinery is damaged or destroyed, the consequences cascade through every sector that depends on its output: gasoline and diesel for transportation and freight, jet fuel for military and commercial aviation, heating oil for residential customers in the Northeast, and feedstocks for the petrochemical production that underlies plastics, fertilizers, and pharmaceuticals.

No facility class in the petroleum sector has longer recovery timelines, more concentrated geographic exposure, or more consequential multi-product output than the refinery. The Gulf Coast cluster — approximately 45 percent of U.S. refining capacity concentrated in a coastal corridor accessible from the Gulf of Mexico — represents both the sector's greatest economic asset and its most significant single-point vulnerability under military-action conditions.

This document establishes the Design Basis Accident scenarios for deliberate military action against petroleum refinery infrastructure. It inherits the threat taxonomy, bounding methodology, and consequence analysis structure from the DBA-OIL Sector Framework. It applies them to the specific physical, operational, and regulatory characteristics of the petroleum refining facility class. The analysis is written for a dual audience: refinery security and operations personnel responsible for facility-level planning, and federal regulators, defense planners, and allied partners responsible for sector-level and alliance-level energy security.

The evidentiary anchor for this document is the Russian campaign against Ukrainian petroleum refining infrastructure, which proceeded systematically from 2022 through 2024 and demonstrated that a state actor with precision-strike capability will target the specific process equipment whose loss maximizes disruption duration, rather than merely the most physically accessible target. The lessons are not Ukraine-specific. They define the Design Basis Threat for any petroleum refinery in the threat environment of a modern peer conflict.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident—Military Action scenarios for the petroleum-refining facility class. Its purpose is to define the bounding threat event that refinery security planning, resilience investments, and recovery architecture must address. As with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

A refinery security program designed against a single-attack-vector threat will fail when a sophisticated adversary integrates cyber pre-positioning, precision kinetic action, and supply chain denial in a coordinated sequence. The DBA-MA approach defines the coordinated, multi-vector attack as the governing design event.

1.2 Scope and Applicability

This document addresses petroleum refinery facilities: large-scale crude oil processing plants that produce gasoline, diesel, jet fuel, heating oil, liquefied petroleum gas, and petrochemical feedstocks through distillation, cracking, treating, and blending. The primary reference environment is the U.S. Gulf Coast refinery cluster, with allied refinery infrastructure as a reference, where the supply chain consequences of military action extend beyond domestic boundaries.

Upstream crude oil production, product pipelines, and bulk liquid storage terminals are addressed in their respective DBA-OIL facility-class documents. Where the refinery's crude supply inputs or product outputs are affected by events in those facility classes, this document addresses the consequence chain as it propagates into the refinery, without displacing the dedicated facility-class analysis for the initiating sector.

1.3 Relationship to Governing Documents

This document is issued under the DBA-OIL Sector Framework (v1.3), which is itself issued under the DBA-MA Parent Framework (IAEA L1 v1.4). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. This facility-class document adds the refinery-specific physical configuration, process unit taxonomy, bounding events, and consequence chains that the sector framework establishes in general terms.

1.4 The Refinery as a Multi-Product Facility

A fundamental characteristic of the refinery facility class distinguishes it from every other facility class in the DBA-OIL series: a refinery attack simultaneously disrupts multiple product streams. A crude oil pipeline attack disrupts crude supply to refineries. A product terminal attack disrupts the distribution of a specific product in a specific region. A refinery attack eliminates production of all products simultaneously — gasoline, diesel, jet fuel, heating oil, and petrochemical feedstocks — from the affected facility. This multi-product consequence is the defining characteristic of refinery attacks and the primary reason the refinery facility class carries the highest consequence weight in the DBA-OIL series.

2. Refinery Profile and Asset Taxonomy

2.1 Refinery Architecture

A petroleum refinery is a continuous process facility that operates without scheduled shutdowns except for planned maintenance turnarounds, typically every three to five years. The core refining process begins with atmospheric distillation: crude oil is heated in a fired heater and fed to a distillation column, where it separates by boiling point into fractions — naphtha, kerosene, gas oil, and residuum. These fractions are further processed in downstream units to produce finished products.

The process unit taxonomy relevant to DBA-MA consequence analysis includes:

Crude Distillation Unit (CDU). The entry point for all crude oil processing. Loss of a CDU shuts down all downstream processing that depends on its output. Most large refineries have one primary CDU; some have two. CDU capacity typically defines the refinery's nameplate crude processing rate.

Fluid Catalytic Cracker (FCC). Converts heavy gas oil fractions into lighter, higher-value products, primarily gasoline and diesel. The FCC is the highest-value process unit in most refineries and is the primary determinant of gasoline production capacity. FCC replacement or major repair requires twelve to eighteen months under normal supply chain conditions.

Hydrocracker. Converts heavy fractions into diesel and jet fuel using hydrogen under high pressure. Refineries configured for high distillate (diesel and jet fuel) production rely heavily on hydrocracker output. Hydrocracker damage is the primary constraint on the recovery of jet fuel production after a military action.

Hydrotreater units. Remove sulfur from product streams to meet fuel specifications. Every product stream requires hydrotreating before distribution. Loss of hydrotreater capacity does not immediately stop refinery operations but can prevent product from meeting sales specifications, effectively idling production even if the primary processing units are intact.

Alkylation unit. Produces high-octane gasoline blending component from light olefins. Alkylation units use hydrofluoric acid (HF) or sulfuric acid as catalysts; HF alkylation units present a unique hazard profile — an HF release creates a toxic vapor cloud that is the refinery's most significant off-site consequence risk. The HF hazard profile makes alkylation units a high-consequence target for adversaries seeking both supply disruption and community consequence.

Hydrogen plant. Produces hydrogen for hydroprocessing operations. Refineries processing sour crude or producing ultra-low-sulfur fuels are highly dependent on hydrogen supply; loss of hydrogen production shuts down the hydroprocessing train, limiting the refinery to producing only unfinished products that cannot be sold.

Fired heaters. Heat crude oil and process streams throughout the refinery. There are typically dozens of fired heaters at a large refinery; each is a potential source of fire and explosion if the

process stream is released at operating temperature. Fired heater damage is a common secondary consequence of explosive events at adjacent process units.

Utilities. Steam generation, cooling water, instrument air, and electrical power supply. Refinery process units cannot operate without utilities. A utility system attack — targeting the power substation, cooling water intake, or steam generation plant — can shut down the entire refinery without directly damaging any process unit.

2.2 Gulf Coast Cluster Profile

The Houston Ship Channel / Beaumont / Port Arthur / Lake Charles refinery corridor represents the highest-concentration refinery cluster in the United States. Approximately fifteen major refineries within a sixty-mile coastal corridor account for approximately twenty percent of total U.S. refining capacity. These facilities are accessible from the Gulf of Mexico through the Ship Channel and the Sabine-Neches Waterway. They share crude supply pipelines, product pipeline interconnections, and marine crude import infrastructure.

The cluster concentration creates both economic efficiencies — shared infrastructure, labor pools, and logistics — and strategic vulnerability. A coordinated attack on three or four major refineries within this corridor, each targeting the highest-consequence process unit with the longest replacement lead time, can produce a sustained reduction in U.S. gasoline, diesel, and jet fuel production capacity that the SPR drawdown can partially but not fully compensate.

2.3 Regulatory Authority

OSHA. The Process Safety Management (PSM) standard under 29 CFR 1910.119 requires process hazard analysis, mechanical integrity programs, and emergency response plans for refineries that handle listed highly hazardous chemicals at or above threshold quantities. PSM is the primary safety regulatory framework for refinery operations and provides the baseline for the non-adversarial Design Basis process safety events.

EPA. The Risk Management Program (RMP) under 40 CFR Part 68 requires worst-case release scenario analysis and emergency response coordination for facilities that handle regulated substances, including HF alkylation units. RMP consequence analysis provides public-domain data on worst-case HF release consequences that are directly relevant to the DBA-MA threat analysis.

TSA. Mandatory cybersecurity directives for petroleum pipeline operators following the Colonial Pipeline incident. Physical security requirements for refineries remain voluntary; TSA has issued chemical facility security guidelines, but they carry no enforcement authority for petroleum refineries.

CISA. The Chemical Facility Anti-Terrorism Standards (CFATS) apply to facilities that handle chemicals of interest at or above threshold quantities, including some refinery chemicals. However, petroleum refining facilities have historically been subject to a DHS exemption from

full CFATS coverage; the scope of that exemption has been contested, and its regulatory status should be verified in current guidance.

3. Design Basis: Petroleum Refining Facility Class

3.1 Non-Adversarial Bounding Events: Refinery

The non-adversarial Design Basis for the petroleum refining facility class is established in the DBA-OIL Sector Framework (Sections 4.2.1 and 4.4.1). The bounding events are major refinery fire and explosion (Section 4.2.1) and control system failure (Section 4.4.1), with the refinery-to-transport cascade (Section 4.5.1) as the systemic bounding event. This document develops the facility-class-specific consequence chains and adequacy standards that the sector framework defines in general terms.

The Texas City refinery explosion (2005) and the Amuay refinery explosion (Venezuela, 2012) are the primary reference events for the non-adversarial fire-and-explosion bounding case. Both demonstrate the chain of consequences from a process unit release through ignition, explosion, and escalation to adjacent units. Both also demonstrate the challenge of the recovery timeline: Texas City required years of reconstruction, and the loss of production capacity affected Gulf Coast product supply for an extended period.

3.2 Design Basis Accident — Military Action: Bounding Event

The DBA-MA bounding event for the petroleum refining facility class is a coordinated, multi-vector attack against a major refinery or multiple refineries within the same regional cluster, executed during a peak demand period, with cyber pre-positioning and kinetic execution integrated to maximize process unit damage and extend recovery timelines.

The bounding parameters, inherited from the sector framework threat taxonomy:

Adversary tier: Tier 1 state actor, with Tier 1 cyber pre-positioning and Tier 2 or direct kinetic execution.

Modality: Cyber-kinetic combined arms. DCS compromise suppresses safety system response and manipulates process conditions to amplify physical damage; kinetic action targets high-consequence process units with maximum replacement lead times.

Target selection: FCC, hydrocracker, and hydrogen plant targeted in sequence to maximize the duration of multi-product supply disruption. HF alkylation unit considered for community consequence amplification—utilities targeted to prevent the restart of undamaged units.

Timing: Peak demand season — winter heating oil demand for Northeast markets or summer driving season for gasoline. Coordinated with elevated European or allied fuel import demand to limit alternative supply routing.

Simultaneity: Attacks on two or more major refineries within the Gulf Coast cluster within a window that exceeds the sector's adaptive capacity — exhausting specialty contractor resources and spare equipment availability simultaneously.

3.3 Dual Design Basis Architecture

Consistent with the DBA-OIL Sector Framework (§5.4) and DBA-EN-SF (§4.2), a petroleum refinery facility may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series (Project 2). The two design bases interact at the DBA level: a postulated physical event may simultaneously stress the AI systems that support facility response, and an AI governance failure may impair the facility's ability to respond to a physical postulated event. The Bright Line and Command Broker architecture established in the DBA-OIL Sector Framework §5.4 and CB-Framework_Cross_Sector v0.2 is the primary mechanism for managing the boundary between the two design bases.

The dual-DB interaction for the petroleum refining facility class is particularly acute because refinery operations are among the most AI-augmented environments in the critical infrastructure sectors: advanced process control (APC) systems, real-time optimization (RTO) engines, and increasingly Gen AI advisory tools operate continuously across every major process unit. The four refinery-specific AI governance interaction points identified below represent the design basis for the AI Governance DB at this facility class.

Process unit optimization and advanced process control. APC and RTO systems that continuously adjust process unit set points — CDU preheat train temperatures, FCC riser temperatures, and catalyst circulation rates, hydrocracker reactor pressures — are ML systems operating below the Bright Line. They are deterministic, formally testable within their training envelopes, and may operate autonomously. However, their set point recommendations directly govern process conditions; an adversarially manipulated APC output that drives a process unit outside its safe operating band is functionally equivalent to direct adversary control of the unit. The Scenario A DCS compromise relies on exactly this pathway: the cyber component manipulates APC or DCS outputs to create conditions that the kinetic component then exploits. The AI Governance Design Basis must establish that APC and RTO systems are architecturally isolated from the corporate IT network, that their outputs are range-checked against independent process safety limits before execution, and that anomalous set point trajectories trigger operator alerts before execution rather than after.

Non-adversarial APC optimization-safety boundary — the structural risk. The paragraph above addresses the adversarial pathway: an adversary manipulating APC outputs to create exploitable conditions. The governing analytical contribution of DBA-GAS-FC2 §3.3 — propagated here per DBA-EN Gaps and Recommendations §3.3 — is that a structurally equivalent risk exists without any adversary. APC and RTO systems at petroleum refineries are commercially incentivized to continuously push process conditions toward the limits that maximize product yield and margin. A crude distillation unit preheat train optimized for maximum throughput operates near its fouling and coking limits. A hydrocracker operating at maximum hydrogen consumption operates near its reactor temperature runaway boundary. A catalytic reformer maximizing octane production operates near its coke laydown and regeneration cycle limits. In each case, the APC system that produces the commercially optimal outcome is simultaneously the system that most consistently tests the safety instrumented system (SIS) trip points that protect against catastrophic process excursion. This is not a hypothetical future risk — it is the operating condition that the EPA Risk Management Program and OSHA Process Safety Management standard were designed to regulate. The AI Governance Design Basis must establish that APC and RTO pre-authorized envelopes include explicit safety margin setbacks from IEC 61511-defined SIS trip points, enforced by architecture rather than by procedure: the APC system must be physically incapable of recommending set points within the exclusion zone around SIS trip points, regardless of the commercial optimization objective. This architectural separation is the Bright Line between commercial optimization and process safety. An APC system deployed without it is operating with its optimization objective and its safety boundary in the same control plane — the condition that DBA-GAS-FC2 §3.3 established as the governing non-adversarial structural risk in continuous-process facilities and that applies with equal force to petroleum refineries. This setback construction is the refining-sector instantiation of FD-BL-D1 R3: a change to the

pre-authorized envelope toward an SIS trip point is excluded from advance discharge and routes to human-authorized change control.

Fired heater management AI advisory tools. Fired heaters are the refinery's most widely distributed high-energy ignition sources, present throughout every major processing train. Gen AI tools for fired heater performance monitoring, burner management advisory output, and heat release optimization are increasingly deployed across refinery operations. These tools are Gen AI — non-deterministic, not formally testable — and operate above the Bright Line. A human Command Broker must be in the control loop for any fired heater operating parameter change recommended by a Gen AI advisory tool. This requirement is not procedural; the control system architecture must enforce it. A Gen AI tool that presents fired heater set point recommendations as pre-populated control system inputs, or that requires the operator to actively reject rather than actively confirm each recommendation, violates the Command Broker architecture regardless of how the tool is characterized by its vendor. The Scenario A progression demonstrates why: the adversary's DCS manipulation targeted the CDU preheat train precisely because fired heater set point manipulation is the highest-consequence pathway available to a cyber actor with DCS access.

Safety Instrumented System Gen AI advisory tools. Safety instrumented systems (SIS) are the last line of automated defense against process excursions that would otherwise result in a release, fire, or explosion. SIS logic is by design deterministic and formally verifiable; it operates below the Bright Line and must remain so. The AI governance concern at the SIS layer is not autonomous SIS operation but the emergence of Gen AI advisory tools that present SIS bypass recommendations, SIS logic modification guidance, or SIS proof-test scheduling optimization. These advisory functions are Gen AI operating above the Bright Line. Any SIS parameter change — including bypass authorization, logic modification, or proof-test interval extension — recommended by a Gen AI tool requires Command Broker authorization, meaning the review and affirmative approval of the qualified SIS engineer responsible for the unit, not merely acknowledgment of an alert. The Scenario A DCS manipulation that suppressed alarms and inhibited ESD actuation is the military-action analog of an adversarially manipulated Gen AI SIS advisory tool. Both pathways produce the same physical consequence through different mechanisms.

Emergency shutdown system ML and the denial characteristic. Emergency shutdown (ESD) systems are deterministic ML systems operating below the Bright Line; they must execute without Command Broker interposition when process conditions exceed defined thresholds. The design basis requirement for ESD under military action conditions is that the system must execute correctly when the DCS environment in which it operates has been compromised. This requires architectural separation of ESD logic from the process DCS. ESD systems must be capable of executing their shutdown functions based on hardwired process inputs, independent of DCS data integrity. The Scenario A consequence envelope demonstrates the failure mode: DCS manipulation that inhibited ESD actuation transformed a manageable process upset into a catastrophic escalation. The AI Governance Design Basis must establish that ESD systems are

architecturally independent of any ML or Gen AI component that processes DCS data, so that adversarial corruption of the DCS layer cannot propagate into ESD logic. Under the denial characteristic of military action, the Command Broker for Gen AI advisory tools supporting emergency response may be unreachable or operating under duress; the design basis must establish minimum response capabilities that remain executable without Gen AI advisory support when communications are degraded.

4. Design Basis Accident Scenarios

4.1 Scenario A: Coordinated Cyber-Kinetic Refinery Attack

4.1.1 Initiating Event

Simultaneous cyber-kinetic attack against a major Gulf Coast refinery. The cyber component, having achieved DCS access through a multi-stage intrusion campaign over the preceding months, initiates the coordinated manipulation of the crude distillation unit preheat train and the FCC reactor-regenerator pressure balance. The DCS manipulation is designed to cause a CDU preheat exchanger tube failure — releasing hot crude into the furnace firebox — while simultaneously driving the FCC into an afterburn condition in the regenerator. Both events would normally be detected and mitigated by operator response and safety systems; the DCS manipulation suppresses the relevant alarms and inhibits ESD actuation for a defined time window.

The kinetic component executes within the alarm-suppression window: precision munitions or UAS-delivered charges against the FCC main column structure and the refinery power substation. The CDU fire, the FCC incident, and the loss of electrical power occurred within minutes of each other.

4.1.2 Progression Sequence

T+0: Simultaneous CDU preheat fire, FCC afterburn, and substation destruction—partial alarm suppression by the cyber component. Emergency shutdown initiated at some subsystems; inhibited at others. Site fire brigade responds to CDU fire.

T+0 to T+30 minutes: CDU fire escalates as the preheat exchanger failure continues to feed crude to the fire source; the FCC afterburn causes refractory damage in the regenerator; cooling water pumps lose power from the substation destruction, degrading fire suppression capability. Operators in the control room are working from partially compromised DCS data.

T+30 minutes to T+4 hours: Mutual aid fire response activated. Loss of cooling water accelerates the escalation of a fire to adjacent process units. The CDU fire threatens the atmospheric vacuum unit immediately downstream. FCC is in emergency shutdown with refractory damage; the extent is unknown until the unit can be opened for inspection. Substation destruction means the entire west processing area is without power, preventing the restart of physically undamaged units.

T+4 to T+72 hours: Major fire knocked down. Damage assessment begins. CDU preheat train destroyed; CDU will require months of repair. FCC refractory damage extent being assessed — initial estimate: six to twelve months for refractory replacement and full mechanical inspection before restart. Substation replacement procurement initiated: lead time assessment indicates 12 to

16 weeks for the main transformer. DCS forensic investigation ongoing; extent of compromise unknown.

T+72 hours onward: Refinery operating at significantly reduced throughput using undamaged secondary distillation units, limited by crude supply routing through undamaged exchangers. Product output across all streams reduced by sixty to seventy percent of nameplate capacity. The FCC refractory repair timeline and substation replacement govern recovery.

4.1.3 Consequence Envelope

Immediate production impact. Loss of FCC and CDU primary operation eliminates approximately 60–70 percent of the facility’s gasoline and diesel production and 100 percent of its FCC-derived products. A refinery with a 300,000-barrel-per-day nameplate capacity produces approximately 130,000 barrels per day of gasoline and 90,000 barrels per day of diesel under normal operations; both are eliminated during the FCC repair.

Multi-product cascade. Jet fuel production is unaffected if the hydrocracker remains undamaged; however, the hydrogen plant’s integrity must be verified before hydroprocessing can resume. Heating oil production is curtailed proportionally with distillation capacity. Petrochemical feedstock production is eliminated with FCC operation.

Recovery timeline. The governing constraint is FCC refractory repair: a minimum of six to twelve months. Substation replacement (twelve to sixteen weeks) gates the restart of undamaged units that cannot operate without power. DCS forensic investigation and reconstitution (four to eight weeks) must be completed before operators can trust any automated safety system response. The compound effect of three overlapping recovery timelines — FCC repair, substation replacement, DCS reconstitution — means the refinery returns to full production only after the longest single-item timeline is complete, with no opportunity to restart undamaged units until all three are resolved.

4.2 Scenario B: Multi-Refinery Cluster Attack

4.2.1 Initiating Event

Coordinated attacks against three major refineries within the Houston Ship Channel / Beaumont / Port Arthur corridor, executed within a 48-hour window. Each attack targets the same unit type — the FCC — using the same modality as Scenario A, with cyber pre-positioning achieved independently at each facility over the preceding months. The simultaneity is the attack’s defining characteristic: it is not three independent events but a single, coordinated campaign designed to exhaust the Gulf Coast’s recovery resources.

4.2.2 The Simultaneity Trap Applied

The Gulf Coast refinery cluster’s resilience mechanisms are designed for sequential, single-facility events. The specialist FCC repair contractors — a small community of firms with the

refractory and mechanical expertise to execute FCC repair — can address one or two simultaneous outages in the region. Three simultaneous FCC outages exceed the combined capacity of the facilities; each facility queues for contractor availability, extending individual recovery timelines beyond what they would be in isolation.

Similarly, the spare FCC components — main column internals, cyclone separators, slide valves — that would be procured for single-facility repair are in finite global supply. Three simultaneous procurement requests for the same components extend delivery timelines for each facility. The adversary who understands this dynamic designs the three-refinery attack not because any single refinery's loss is insufficient, but because the simultaneous attack exhausts the recovery system's capacity at every level.

4.2.3 Consequence Envelope: Cluster Attack

Supply impact. Three major Gulf Coast refineries representing approximately eight to ten percent of total U.S. refining capacity are offline simultaneously for six to twelve months. Gasoline and diesel production reduction of approximately 800,000 to 1,000,000 barrels per day from the affected facilities. SPR drawdown provides crude supply continuity but does not substitute for refinery processing capacity — crude oil in the SPR cannot be converted to gasoline or diesel without functioning refineries.

National market consequence. Gasoline and diesel price increases spread nationally within days of the attack's confirmation. Regional shortfalls on the Eastern Seaboard, which is heavily dependent on Gulf Coast product pipeline supply, develop within one to two weeks. Aviation fuel supply for Gulf Coast military installations is immediately affected; jet fuel allocation prioritization for military consumers begins within the first 72 hours.

Allied supply consequence. U.S. petroleum product export capacity — primarily diesel and jet fuel to European and Latin American buyers — is curtailed. Allied nations with supply commitments dependent on U.S. Gulf Coast product exports face supply stress simultaneously with the military threat that motivated the attack.

4.3 Scenario C: HF Alkylation Unit Attack

4.3.1 Initiating Event

Targeted kinetic attack against the hydrofluoric acid alkylation unit at a major Gulf Coast refinery, designed to achieve a large HF release. The HF alkylation unit is targeted not for its contribution to refinery supply output — alkylate production is significant but not the refinery's most critical product stream — but for its off-site consequence potential. HF is an acutely toxic substance; a large release creates a toxic vapor cloud whose downwind hazard zone can extend for miles from the release point, affecting the surrounding community and creating a mass-casualty event that compels emergency response resource allocation and generates public and political pressure that is the adversary's secondary strategic objective.

4.3.2 HF Release Consequence Envelope

Immediate on-site consequences. Injuries and fatalities among refinery personnel in or near the alkylation unit. Emergency evacuation of the refinery. Site emergency response focused on HF cloud containment rather than process unit firefighting.

Off-site community consequences. Downwind HF vapor cloud extending potentially one to three miles from the release point under stable atmospheric conditions, depending on release rate and quantity. Emergency shelter-in-place or evacuation orders for the surrounding community. The hospital system is surging from HF exposure casualties. Long-term HF exposure effects — lung damage, systemic fluoride toxicity — require extended medical follow-up for exposed individuals. The surrounding community consequence is the attack's primary strategic effect: a visible mass-casualty event adjacent to a major industrial facility, creating public and political pressure that exceeds the impact of the supply disruption alone.

Regulatory consequence. OSHA and EPA investigations. Potential shutdown of HF alkylation operations nationally pending safety review. The 2019 Philadelphia Energy Solutions refinery fire, which involved an HF alkylation unit, produced exactly this regulatory dynamic on a smaller scale: OSHA and EPA investigations, community pressure, and political scrutiny of HF alkylation operations nationwide. A deliberate, large-scale HF release would produce a regulatory response of far greater scale and duration.

Note on risk communication. The HF alkylation unit attack scenario raises specific risk communication challenges for DBA-MA document distribution. The consequence analysis in this section is based on public-domain EPA RMP data and published process-safety literature. The analytical purpose is protective: identifying the highest-consequence target in the refinery for priority hardening investment. This posture is consistent with the nuclear safety community's treatment of spent fuel pool vulnerability analysis.

5. Protective Action Thresholds

5.1 Threat Detection and Indicator Taxonomy

5.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary interest in Gulf Coast refinery infrastructure or specific refinery facility layouts. Anomalous DCS network activity: unusual process historian queries, lateral movement between corporate and OT network segments, anomalous vendor access sessions outside maintenance windows. Physical surveillance of refinery fence lines or marine access points. Unusual aerial UAS activity in refinery airspace or adjacent waterway corridors. Procurement inquiries for refinery process engineering data or specific equipment specifications.

5.1.2 Imminent Threat Indicators

Confirmed unauthorized aerial UAS in refinery airspace. Simultaneous DCS anomalies at multiple refineries within the Gulf Coast cluster. Confirmed physical intrusion at refinery perimeter. Simultaneous alarm suppression events at process units at two or more facilities within a short time window — a pattern inconsistent with coincidental equipment failure. Confirmed kinetic strike on any refinery facility within the cluster.

5.2 Decision Authority Framework

Facility level: Refinery operator authority to initiate emergency shutdown, process unit isolation, and site evacuation. TSA and CISA notification is required for confirmed or suspected cyber incidents affecting ICS systems. Local emergency management notification for events with potential off-site consequences (HF release, large fire).

Federal coordination: CISA coordinates federal cyber incident response. DOE ESF-12 activation for supply disruption response. EPA and USCG Federal On-Scene Coordinator activation for environmental releases. OSHA incident response for process-safety events involving fatalities or serious injuries.

Defense coordination: For events with confirmed or strongly suspected state-actor attribution, National Security Council coordination and DOD involvement in response planning.

5.3 Federal Coordination Thresholds

Confirmed kinetic attack on any major Gulf Coast refinery process unit. Confirmed DCS compromise with suspected nation-state attribution at any petroleum refinery. Simultaneous security incidents at two or more refineries within the Gulf Coast cluster within 48 hours. HF release affecting or threatening to affect the surrounding community. Supply disruption affecting military logistics fuel contracts above the defined threshold volumes.

6. Recovery Architecture

6.1 Refinery Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Emergency shutdown and fire suppression. Site security and perimeter control. DCS forensic investigation initiation. Military fuel reserve activation and priority allocation assessment. Product pipeline emergency routing assessment for supply from undamaged facilities. SPR drawdown authorization for crude supply emergency.

Phase 2 — Short-term (72 hours to 8 weeks). Damage assessment completion. DCS forensic investigation complete; system reconstitution underway. Undamaged process units restarted as utilities are restored and DCS integrity confirmed—specialty contractor mobilization for damaged units. Replacement equipment procurement initiated. Alternative product supply routing from unaffected regions maximized.

Phase 3 — Medium-term (8 weeks to recovery). FCC refractory repair and mechanical inspection. Process unit reconstruction where required. Safety case re-validation and regulatory inspection. Sequential restart following regulatory clearance. Supply chain monitoring for adversary interdiction of replacement equipment.

6.2 Critical Recovery Constraints

FCC repair lead time. The fluid catalytic cracker is the governing recovery constraint for the Scenario A and B consequence envelopes. Refractory replacement requires scaffolding, refractory demolition, new refractory installation, curing, and mechanical inspection before restart — a minimum of six months for a major refractory repair, typically eight to twelve months with regulatory inspection and restart. No shortcut exists; the refractory must cure at the temperature before the unit can be operated.

Specialty contractor capacity. FCC repair requires specialist contractors with expertise in refractory and process units. In Scenario B, the cluster attack, three simultaneous FCC outages exceed the available capacity of specialist contractors in the Gulf Coast market, extending each facility's recovery timeline because contractors are allocated sequentially. Pre-established contractor reservation agreements and national contractor capacity tracking are the mitigation mechanisms; neither currently exists in any mandatory framework.

Power infrastructure. The replacement of the main transformer governs the restart of all electrically dependent systems in the affected process areas. Transformer lead times of 12 to 16 weeks under normal supply chain conditions can be extended if the design-basis attack has interdicted the supply chain for replacement equipment. The electricity sector's large power transformer (LPT) spare equipment program serves as a model; no equivalent program exists for the petroleum sector's electrical infrastructure.

DCS reconstitution. Refinery DCS forensic investigation and reconstitution must be completed before any automated safety system can be trusted for process restart. This timeline — a minimum of four to eight weeks — runs in parallel with physical repair but must be completed before the restart, regardless of the physical repair status. A refinery whose process units are physically repaired cannot restart until DCS integrity is confirmed.

6.3 Supply Chain Monitoring

The adversary who executes a Scenario A or B attack understands the recovery critical path. Supply chain interdiction targeting FCC refractory materials, main transformers, and DCS hardware — components with long lead times and concentrated supply chains — can substantially extend the effective disruption period beyond the physical damage timeline. Recovery planning must include supply chain monitoring for signs of adversary interdiction and pre-positioned procurement of critical long-lead-time items before an attack occurs.

7. Gaps and Recommendations

7.1 Regulatory Gap: Physical Security Standards

Petroleum refineries have no mandatory physical security standard. CFATS covers some refinery chemicals; TSA security guidelines are voluntary for physical security. A facility that processes hundreds of thousands of barrels per day of crude oil, maintains a hydrofluoric acid alkylation unit with an off-site consequence radius of several miles, and whose loss would affect national fuel supply for months, has no regulatory requirement to assess its vulnerability to the Design Basis Threat defined in this document. This is the most consequential single gap in the U.S. petroleum sector security framework.

7.2 HF Alkylation Unit Risk Mitigation

The HF alkylation unit attack scenario illustrates a risk that the petroleum industry has managed for decades without resolving: the use of large quantities of hydrofluoric acid in refinery operations near populated areas. Modified HF alkylation processes that use less mobile HF or alternative catalysts (sulfuric acid alkylation, solid acid processes) exist and have been adopted at some facilities. A DBA-MA informed security assessment of HF alkylation risks should inform both facility hardening investment and technology conversion decisions. The regulatory framework for this decision should be developed before a military action event creates the political conditions for reactive, poorly designed regulation.

7.3 Cluster Resilience Planning

The Gulf Coast refinery cluster lacks a system-level resilience plan that addresses the consequences of a simultaneous attack under Scenario B. Specialty contractor capacity, spare equipment pre-positioning, and alternative product supply routing for a multi-facility cluster event have not been analyzed at the system level by any federal agency with the authority and the data access to do so. DOE, in coordination with TSA, CISA, and the American Fuel and Petrochemical Manufacturers (AFPM), should conduct a cluster resilience assessment using the Scenario B consequence envelope as the design basis.

7.4 DCS Security and Anomaly Detection

Refinery DCS systems are the primary pre-attack detection opportunity for the Scenario A and B cyber component. ICS anomaly detection capable of identifying the behavioral signatures of DCS pre-positioning — unusual historian access, anomalous vendor sessions, and lateral movement from IT to OT networks — is commercially available and has been deployed at some facilities. No regulatory requirement mandates its deployment at petroleum refineries. TSA's mandatory cybersecurity directives for pipeline operators provide a model; the same framework should be extended to cover petroleum refinery control systems.

Revision History

Version	Date	Author	Description
1.0	April 2026	Tim Roxey	Initial version. Establishes Design Basis Accident — Military Action scenarios for the petroleum refining facility class. Issued concurrently with DBA-OIL Sector Framework v1.0. Three scenarios were developed: Scenario A (coordinated cyber-kinetic single-refinery attack), Scenario B (multi-refinery Gulf Coast cluster attack), Scenario C (HF alkylation unit attack). Recovery constraints, cluster resilience gap, and regulatory gaps were identified.
1.1	June 2026	T. Roxey	Section 3.3 (Dual Design Basis Architecture) added. Establishes four refinery-specific AI governance interaction points: process unit optimization and APC (below Bright Line, autonomous permitted with isolation and range-check requirements); fired heater management Gen AI advisory tools (above Bright Line, Command Broker required, pre-populated inputs prohibited); SIS Gen AI advisory tools (above Bright Line, qualified SIS engineer as Command Broker); and ESD system ML and the denial characteristic (deterministic, Bright Line-independent, architectural isolation from DCS required). Section 5.3 renamed from “Bright Line Criteria” to “Federal Coordination Thresholds”; “Bright Line” reserved exclusively for the ML/AI governance boundary throughout this document. Resolves the dual-DB gap identified in DBA-EN-SF §4.2 and Priority 2 of the architecture review. The governing foundational instruments are FD-BL (placement — consequence, per action), FD-BL-D1 (mode of discharge), FD-BL-D2 (indication integrity), and FD-EV (envelope), applied by reference; per FD-EV, the monitored operating-limit sense of “envelope” is the operating band and the pre-authorized/setback sense is the pre-authorized envelope.
1.2	June 2026	T. Roxey	Housekeeping revision. Parent document reference updated from DBA-OIL Sector Framework v1.0 to v1.3 on cover, §1.3, and cover lineage line. “Brightline” corrected to “Bright Line” (two words) throughout §3.3 and the v1.1 revision history entry, consistent with series-wide convention. v1.1 revision history entry cleaned up to remove erroneous OI-1 reference; description now accurately reflects the §3.3 addition and Section 5.3 renaming performed in v1.1. No substantive content changes.

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